

Problem A. PDAO Game

Input file: stdin
Output file: stdout
Time limit: 1 second
Memory limit: 256 MB

You are now playing a VRMMORPG game called **PDAO (Programming Design Art Online)**. The game features a floating castle named *IMcrad* with 100 floors. As part of the raiding team, you must conquer its stages. Now, in the game, you and your teammates are about to challenge a mysterious dungeon filled with traps and mechanisms.

The dungeon is represented as an $n \times n$ grid, where n is an odd integer. Each cell in the grid is one of the following:

- **w**, representing a wall that cannot be passed;
- **-**, representing a road that can be passed;
- **0**, representing the treasure room.

There is exactly one treasure room in the dungeon, located at position (t_1, t_2) .

At the entrance of the dungeon lies an ancient control chamber. By activating it, you may toggle **an entire row or an entire column** of the dungeon map. When a row or column is toggled, every wall (**w**) becomes a road (**-**), and every road (**-**) becomes a wall (**w**). The treasure cell **0** is not affected by toggling.

You start at position $(0, 0)$ and must first reach the treasure room at (t_1, t_2) . After collecting the treasure, you must escape from position $(n - 1, n - 1)$.

You may move only in the four cardinal directions (up, down, left, right). Once you step on a road cell, it collapses immediately and cannot be used again.

All toggles must be completed **before entering the dungeon**, and no further toggling is allowed afterward. All indices mentioned in this problem are **0-based**.

Determine whether there exists a sequence of row and column toggles that allows you to complete the dungeon successfully.

Input

The first line contains three integers:

n t1 t2

where $n \leq 7000$ is the size of the dungeon grid and (t_1, t_2) is the location of the treasure room.

The next n lines each contain a string of length n , describing the dungeon layout. Each character is one of **w**, **-**, or **0**. The cell at position (t_1, t_2) is guaranteed to be **0**.

It is guaranteed that the treasure is not located on the boundary, i.e., $0 < t_1 < n - 1$ and $0 < t_2 < n - 1$.

Output

Print a single line:

Yes

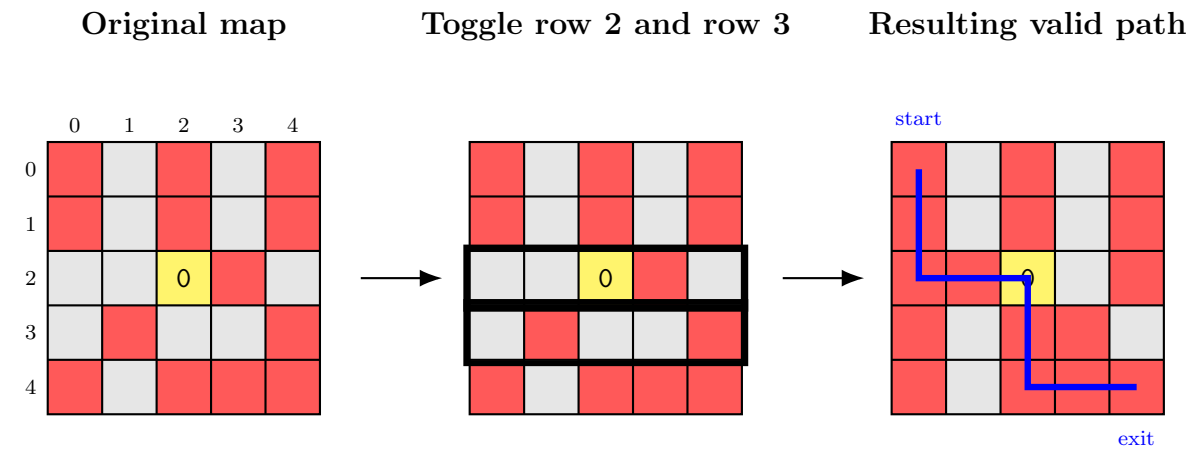
if it is possible to reach the treasure and then escape the dungeon; otherwise, print:

No

Example

stdin	stdout
5 2 2 -w-w- -w-w- ww0-w w-ww- -w----	Yes

Explanation



For the sample input, toggling row 2 and row 3 (0-based) transforms the grid into one that contains a simple valid route

$(0,0) \rightarrow (1,0) \rightarrow (2,0) \rightarrow (2,1) \rightarrow (t_1,t_2) = (2,2) \rightarrow (3,2) \rightarrow (4,2) \rightarrow (4,3) \rightarrow (4,4).$

Hence the answer is **Yes**.